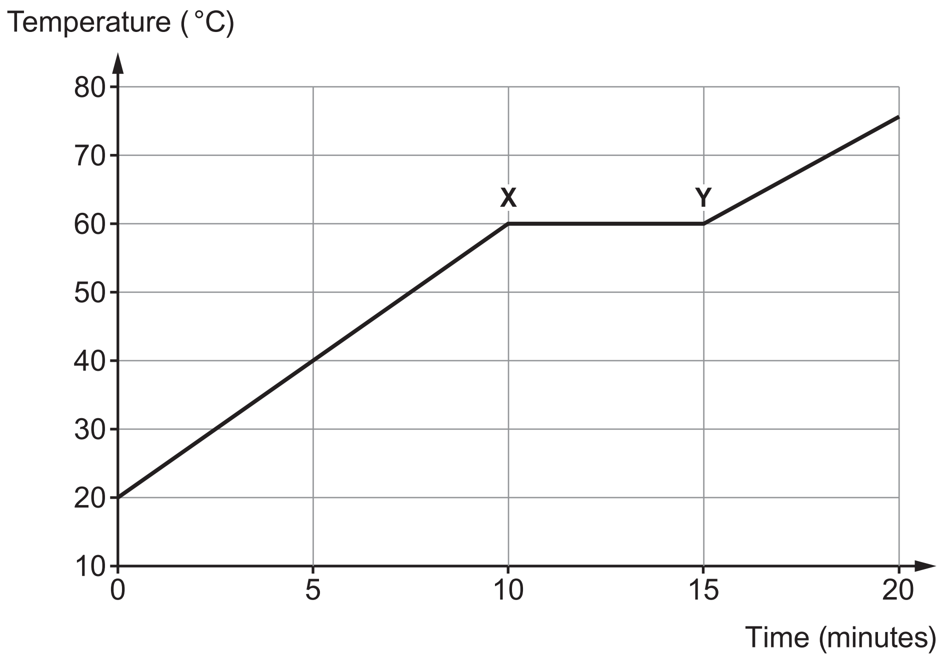
Examiner only

6. The graph below shows the results obtained when **36 g of solid wax** was heated using a heater. The temperature of the wax was recorded every minute.



(a) Explain why the energy supplied from the heater between **points X and Y** doesn't produce a rise in temperature of the wax. [2]

....

....

....

(b) (i) Use the equation:

thermal energy supplied = mass × specific latent heat

to calculate the amount of thermal energy supplied to the 36 g of wax between **points X and Y**. [2]
(Specific latent heat of fusion (L_f) for wax = 220 J/g)

Thermal energy supplied = ... J

Examiner only

(ii) All the energy is transferred from the heater to the wax during the 5 minute interval between **X and Y**.

I. Calculate the time, in seconds, of the interval between **X and Y**. [1]

Time = ... s

II. Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

to calculate the power of the heater. [2]

Power of heater = ... W

7